

■ Nmap Scan Report

Network Scanning & Service Enumeration

Ethical Hacking Internship — Task 02

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Organization	White Band Associates
Task	Task 02 — Scanning and Enumeration
Scan Date	June 2026
Target	localhost / 127.0.0.1
OS Detected	Linux (kernel 5.0 – 6.2)
Total Open Ports	2

■ **Disclaimer:** All scans in this report were performed only on **localhost** (the intern's own machine). No public or unauthorized systems were scanned. This report is prepared solely for educational purposes as part of the White Band Associates Ethical Hacking Internship Programme.

Part A — Installing Nmap

Nmap (Network Mapper) is a free, open-source tool used by security professionals to discover hosts and services on a network. It is the industry-standard first step in any authorised network audit.

Installation (Linux / Ubuntu)

```
sudo apt update sudo apt install nmap -y
```

Verification Command

```
nmap --version
```

Output Received

```
Nmap version 7.98 ( https://nmap.org ) Platform: x86_64-pc-linux-gnu Compiled with:  
liblua-5.4.8 openssl-3.5.5 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6  
Available nsock engines: epoll poll select
```

■ Screenshot saved: screenshots/part_A/Nmap version.PNG

Part B — Scanning My Local Machine

The local machine was scanned using the loopback address. This is a completely safe scan because it only targets the intern's own computer.

Command Used

```
nmap localhost # or nmap 127.0.0.1
```

■ localhost and 127.0.0.1 both refer to the local machine. No external network traffic is generated.

Output Received

```
PORT STATE SERVICE 22/tcp open ssh 80/tcp open http Nmap done: 1 IP address (1 host up)  
scanned in 0.11 seconds
```

Findings Summary

Detail	Value
Total Open Ports	2
Port Numbers	22, 80
Services Running	SSH, HTTP

■ Screenshot saved: screenshots/part_B/localhost.PNG

Part C — Service Version Detection

The -sV flag instructs Nmap to probe each open port and identify the exact software name and version number. Knowing version numbers is critical because older versions often contain publicly-known vulnerabilities.

Command Used

```
nmap -sV localhost
```

Output Received

```
PORT STATE SERVICE VERSION 22/tcp open ssh OpenSSH 10.2p1 Debian 5 (protocol 2.0) 80/tcp  
open http Apache httpd 2.4.66 ((Debian)) Service Info: OS: Linux; CPE:
```

```
cpe:/o:linux:linux_kernel
```

Service Version Table

Port	Service	Version
22	SSH	OpenSSH 10.2p1 Debian 5
80	HTTP	Apache httpd 2.4.66 (Debian)

■ Screenshot saved: screenshots/part_C/Nmap -sV localhost.PNG

Part D — Operating System Detection

The `-O` flag enables OS fingerprinting. Nmap analyses low-level network responses to determine the likely operating system. `sudo` (root) privileges are required because raw packet crafting needs elevated permissions.

Command Used

```
sudo nmap -O localhost
```

Output Received

```
OS details: Linux 2.6.32, Linux 5.0 - 6.2 Network Distance: 0 hops
```

Q&A

Q: Was the OS detected?

A: Yes. Nmap successfully identified the operating system.

Q: Which OS was identified?

A: Linux — kernel range 2.6.32 / 5.0 – 6.2 (Debian-based system).

Q: Why is OS detection useful in penetration testing?

A: Knowing the OS allows testers to target OS-specific vulnerabilities. Some exploits work only on Windows; others only on Linux. It also reveals whether the system is running an outdated, unsupported kernel that may have unpatched CVEs.

■ Screenshot saved: screenshots/part_D/Nmap -O localhost.PNG

Part E — Common Port Research

The table below documents the most commonly encountered network ports, their purposes, and the typical security risks associated with each.

Port	Protocol	Purpose	Security Risks
20/21	FTP	Transfer files between computers	Plain-text transfer; credentials easily intercepted
22	SSH	Secure encrypted remote login	Brute-force attacks with weak passwords
23	Telnet	Legacy unencrypted remote login	All data in plain text — easily sniffed; deprecated
25	SMTP	Sending emails between servers	Open relay abuse; spam propagation
53	DNS	Resolves domain names to IP addresses	DNS poisoning; DDoS amplification

Port	Protocol	Purpose	Security Risks
80	HTTP	Standard unencrypted web traffic	Man-in-the-middle attacks; data exposed in transit
110	POP3	Downloading emails from a server	Plain-text credentials; should use POP3S (995)
143	IMAP	Syncing email across devices	Same risks as POP3; use IMAPS (993) instead
443	HTTPS	Encrypted web traffic (TLS/SSL)	Expired or mis-configured certificates; weak cipher suites
445	SMB	Windows file and printer sharing	EternalBlue / WannaCry ransomware; high risk if unpatched
3389	RDP	Remote desktop access to Windows	BlueKeep (CVE-2019-0708); brute force; exposed to internet

Key Takeaway: Ports that transmit data without encryption (21, 23, 80, 110) are significantly more dangerous than their encrypted counterparts (22, 443, 993, 995).

Part F — Scan Analysis

The following analysis is based on the scan results obtained in Parts B, C, and D.

1. Which services are currently running?

- SSH (Port 22) — provides secure encrypted remote access to the machine.
- HTTP (Port 80) — Apache web server is actively listening for web requests.

2. Are all open ports necessary?

- Port 22 (SSH): Useful only if remote access is genuinely needed; otherwise it should be disabled to reduce the attack surface.
- Port 80 (HTTP): No website is being intentionally hosted on this machine, so the Apache server appears to be running unnecessarily.

3. Which services could become security risks if misconfigured?

- Port 80 (HTTP): An outdated or poorly configured Apache instance can be exploited via directory traversal, remote code execution, or default credentials.
- Port 22 (SSH): Without key-based authentication or fail2ban, SSH is vulnerable to automated brute-force attacks.

4. Which port would you disable if not required?

- Port 80 (HTTP) would be the first to disable. Since no web service is intentionally being hosted, a running web server is an unnecessary risk. SSH (Port 22) would remain — but hardened with key-based login and a non-default port if possible.

Part G — Scan Report Summary

Scan Metadata

Scan Date	June 2026
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Target System	localhost / 127.0.0.1
Operating System	Linux (kernel 5.0 – 6.2, Debian-based)
Total Open Ports	2
Commands Used	nmap localhost nmap -sV localhost sudo nmap -O localhost

Open Ports & Running Services

Port	Service	Version	Status
22	SSH	OpenSSH 10.2p1 Debian 5	Open
80	HTTP	Apache httpd 2.4.66 (Debian)	Open

Observations

- Only 2 ports were found open — a small attack surface, but both require attention.
- Service version detection succeeded; version numbers are clearly identifiable.
- The OS fingerprint matched Linux kernel 5.0 – 6.2, confirming a modern Debian system.
- Port 80 (HTTP / Apache) is running despite no active website being hosted.
- OpenSSH 10.2p1 is the latest major release — this is a positive security indicator.

Recommendations

Recommendation	Priority	Action
Disable Port 80	HIGH	Stop the Apache service if no web hosting is needed.
Harden SSH (Port 22)	HIGH	Enforce key-based auth; disable root login; consider non-default port.
Keep software updated	MEDIUM	Regularly update OpenSSH and Apache to patch new CVEs.
Audit ports periodically	MEDIUM	Run nmap scans regularly to catch unexpected open ports.
Enable a firewall (UFW)	HIGH	Block all inbound ports except those explicitly required.

Conclusion

In this task, I learned how to use Nmap to perform structured network scanning on my own machine as part of the Scanning and Enumeration phase of Ethical Hacking. By running three targeted scans — a basic port scan, a service-version scan, and an OS-detection scan — I was able to build a complete picture of what is running on the system and, crucially, how that information might be used by an attacker.

The experience highlighted that even a personal computer can expose services that are not strictly necessary. Every open port is a potential entry point. Understanding which services are running — and whether they are actually needed — is the first step toward reducing a system's attack surface. Version detection is especially valuable because outdated software versions often carry well-documented CVEs that adversaries actively exploit.

The scanning phase is not an attack in itself; it is reconnaissance that informs every decision that follows in a security assessment. Before we can protect a system, we must see it the same way a potential attacker would. This task gave me that perspective.